

GrounUp Enterprise Specification

Phase 17 - Machine Control, GPS & Field Automation

Version 1.0.0 | Review-ready | Proposed lock: GES-P17-v1.0.0

1. Mission

Create a governed machine-control and field-automation platform that converts approved survey/design data into controlled in-cab guidance and automation while preserving localization, revision, safety, operator, device, production and audit context.

2. Architecture Principles

Approved-model authority; localization integrity; vendor neutrality; explicit machine/device compatibility; operator authority; safety-first automation; resilient sync; exact job/session lineage; auditable overrides; human-governed AI.

3. Phase 16 Handoff

Phase 17 consumes approved surfaces, alignments, control/localization references, CRS/datum/geoid/units, checksums and validation evidence from Phase 16. The handoff is validated before machine job packaging.

4. Machine Job Governance

Every machine job package has a revision, checksum, approved source model/localization, assigned machines/classes, compatibility result, issue status and acknowledgement history. Stale or withdrawn jobs are visibly blocked or warned according to policy.

5. Guidance & Automatic Control

The platform supports 2D and 3D guidance plus governed automatic grade-control integration. Automatic-control enablement requires valid job/localization, healthy positioning, compatible equipment and configured operator/safety prerequisites.

6. Equipment-Specific Workflows

Excavators, dozers, graders, compactors, loaders, pavers, milling machines and trench/utility operations receive specialized guidance, production and quality records while sharing common governance.

7. Field Synchronization & Telemetry

Cloud-to-machine job delivery and machine-to-cloud telemetry/production support intermittent connectivity, resumable transfers, offline cache, acknowledgement and conflict handling.

8. Safety & Revision Control

Geofences, exclusion/no-go zones, position-quality alerts, design revision alerts and operator controls are explicit. AI cannot override safety boundaries or autonomously enable control.

9. Production, As-Built & Quality

Machine sessions can capture cut/fill, passes, conformance, production quantities, material movement and as-built geometry. Records retain exact machine, operator, model, localization, device and session lineage.

10. AI Machine Intelligence

AI may identify anomalies, assist diagnostics, review model/data quality and recommend operational improvements. Consequential changes remain human approved and AI outputs retain evidence/confidence.

Functional Domains

Machine Control Core

Governed machine-control projects, files, revisions, assignments, status and audit.

GPS/GNSS Positioning

Machine GNSS receivers, correction services, localization, positioning quality and health.

Localization & Calibration

Site calibration/localization, control points, residuals, validation, approvals and rollback.

3D Models & Surfaces

Machine-ready surfaces, alignments, corridors, linework, grades, slope models and version control.

2D Guidance

Horizontal/vertical guidance, line/offset, depth, cross-slope and simple grade control.

3D Guidance

Real-time design comparison, cut/fill, blade/bucket position, grade, slope and tolerance.

Automatic Grade Control

Automated blade/bucket/implement control, limits, enablement, intervention and operator authority.

Excavator Guidance

Bucket tip, depth, slope, trench, pipe, bench, reach and avoidance guidance.

Dozer Guidance

Blade elevation, cross-slope, cut/fill, pass tracking, surface conformance and productivity.

Grader Guidance

Moldboard position, cross-slope, design line, surface conformance and finish grading.

Compactor Guidance

Pass count, coverage, speed, amplitude/frequency references, target zones and compaction evidence.

Loader & Material Handling

Stockpile/bench guidance, payload references, zones, haul destinations and production context.

Paver/Milling Guidance

Stringless references, elevation/slope targets, milling depth, paving control and QA interfaces.

Trenching & Utilities

Utility alignment, trench depth, grade, bedding, pipe invert, exclusion zones and as-built capture.

Road & Highway Machine Control

Alignments, stationing, cross-sections, corridors, superelevation and DOT model references.

Site Models & Job Files

Job-file creation, packaging, validation, delivery, device import and revision acknowledgement.

Field Data Sync

Machine-to-cloud, cloud-to-machine, offline sync, conflict detection, resumable transfer and telemetry.

Machine Telemetry Integration

Hours, position, utilization, idle, fuel, fault codes and machine-control activity linkage.

Operator Interface

In-cab UX, alerts, revision status, quality indicators, operator acknowledgement and training mode.

Safety & Exclusion Controls

Geofences, no-go zones, utilities, overhead hazards, slope limits and alert/escalation rules.

Design Change Management

Changed surfaces/alignments, impact analysis, withdrawal, reissue and field acknowledgement.

As-Built & Production Capture

Machine-measured elevations, passes, cut/fill progress, material movement and evidence.

Quality Assurance

Tolerance checks, conformance, checkpoints, calibration status, model validation and exception handling.

Field Automation

Automated workflows, digital work instructions, task sequencing, triggers, alerts and completion evidence.

Remote Support & Diagnostics

Remote session controls, troubleshooting, software/firmware status, logs and support evidence.

Device & Firmware Management

Controllers, displays, sensors, antennas, firmware, compatibility, configuration and release control.

Vendor/Platform Interoperability

Trimble, Topcon, Leica and other vendor-neutral adapters, formats and mapping layers.

AI Machine Intelligence

Evidence-grounded anomaly detection, optimization recommendations, diagnostics and model QA.

Machine Control Digital Twin

Current model, localization, device, operator, machine, telemetry, production and quality state.

Controlled Artifact Summary

Artifact	Count
Requirements	380
Entities	175
Fields	280
Business Rules	75
Validation Rules	75
Workflows	51
Engines	55
Api Resources	68
Ui Surfaces	49
Reports	40
Kpis	51
Tests	380

KPI Registry

P17-KPI-001 - Approved Model Compliance - Machine Control Core - Governed metric definition for Approved Model Compliance.

P17-KPI-002 - Current Job Revision Compliance - GPS/GNSS Positioning - Governed metric definition for Current Job Revision Compliance.

P17-KPI-003 - Localization Verification % - Localization & Calibration - Governed metric definition for Localization Verification %.

P17-KPI-004 - Calibration Residual Pass Rate - 3D Models & Surfaces - Governed metric definition for Calibration Residual Pass Rate.

P17-KPI-005 - CRS/Datum Metadata Completeness - 2D Guidance - Governed metric definition for CRS/Datum Metadata Completeness.

P17-KPI-006 - Device Compatibility Pass Rate - 3D Guidance - Governed metric definition for Device Compatibility Pass Rate.

P17-KPI-007 - Firmware Compliance % - Automatic Grade Control - Governed metric definition for Firmware Compliance %.

P17-KPI-008 - Operator Authorization % - Excavator Guidance - Governed metric definition for Operator Authorization %.

P17-KPI-009 - Job Acknowledgement % - Dozer Guidance - Governed metric definition for Job Acknowledgement %.

P17-KPI-010 - Position Quality Pass Rate - Grader Guidance - Governed metric definition for Position Quality Pass Rate.

P17-KPI-011 - RTK/Correction Availability - Compactor Guidance - Governed metric definition for RTK/Correction Availability.

P17-KPI-012 - Guidance Session Availability - Loader & Material Handling - Governed metric definition for Guidance Session Availability.

P17-KPI-013 - Auto-Control Utilization - Paver/Milling Guidance - Governed metric definition for Auto-Control Utilization.

P17-KPI-014 - Manual Override Rate - Trenching & Utilities - Governed metric definition for Manual Override Rate.

P17-KPI-015 - Cut/Fill Accuracy - Road & Highway Machine Control - Governed metric definition for Cut/Fill Accuracy.

P17-KPI-016 - Finish Grade Conformance - Site Models & Job Files - Governed metric definition for Finish Grade Conformance.

P17-KPI-017 - Cross-Slope Conformance - Field Data Sync - Governed metric definition for Cross-Slope Conformance.

P17-KPI-018 - Excavator Depth Accuracy - Machine Telemetry Integration - Governed metric definition for Excavator Depth Accuracy.

P17-KPI-019 - Dozer Grade Accuracy - Operator Interface - Governed metric definition for Dozer Grade Accuracy.

P17-KPI-020 - Grader Finish Accuracy - Safety & Exclusion Controls - Governed metric definition for Grader Finish Accuracy.

P17-KPI-021 - Compactor Coverage % - Design Change Management - Governed metric definition for Compactor Coverage %.

P17-KPI-022 - Compaction Target Pass Rate - As-Built & Production Capture - Governed metric definition for Compaction Target Pass Rate.

P17-KPI-023 - Paving Elevation Compliance - Quality Assurance - Governed metric definition for Paving Elevation Compliance.

P17-KPI-024 - Milling Depth Compliance - Field Automation - Governed metric definition for Milling Depth Compliance.

P17-KPI-025 - Trench Depth Compliance - Remote Support & Diagnostics - Governed metric definition for Trench Depth Compliance.

P17-KPI-026 - Pipe Invert Compliance - Device & Firmware Management - Governed metric definition for Pipe Invert Compliance.

P17-KPI-027 - Geofence Violation Count - Vendor/Platform Interoperability - Governed metric definition for Geofence Violation Count.

P17-KPI-028 - No-Go Alert Closure - AI Machine Intelligence - Governed metric definition for No-Go Alert Closure.

P17-KPI-029 - Safety Alert Closure - Machine Control Digital Twin - Governed metric definition for Safety Alert Closure.

P17-KPI-030 - Revision Alert Closure - Machine Control Core - Governed metric definition for Revision Alert Closure.

P17-KPI-031 - Sync Success Rate - GPS/GNSS Positioning - Governed metric definition for Sync Success Rate.

P17-KPI-032 - Offline Conflict Rate - Localization & Calibration - Governed metric definition for Offline Conflict Rate.

P17-KPI-033 - As-Built Capture % - 3D Models & Surfaces - Governed metric definition for As-Built Capture %.

P17-KPI-034 - Production Capture Completeness - 2D Guidance - Governed metric definition for Production Capture Completeness.

P17-KPI-035 - Material Movement Accuracy - 3D Guidance - Governed metric definition for Material Movement Accuracy.

P17-KPI-036 - Design Change Acknowledgement Time - Automatic Grade Control - Governed metric definition for Design Change Acknowledgement Time.

P17-KPI-037 - Withdrawal Compliance - Excavator Guidance - Governed metric definition for Withdrawal Compliance.

P17-KPI-038 - Reissue Cycle Time - Dozer Guidance - Governed metric definition for Reissue Cycle Time.

P17-KPI-039 - Device Health % - Grader Guidance - Governed metric definition for Device Health %.

P17-KPI-040 - Remote Support Resolution Time - Compactor Guidance - Governed metric definition for Remote Support Resolution Time.

P17-KPI-041 - Vendor Conversion Success - Loader & Material Handling - Governed metric definition for Vendor Conversion Success.

P17-KPI-042 - Field Automation Completion - Paver/Milling Guidance - Governed metric definition for Field Automation Completion.

P17-KPI-043 - Model QA Pass Rate - Trenching & Utilities - Governed metric definition for Model QA Pass Rate.

P17-KPI-044 - AI Anomaly Acceptance - Road & Highway Machine Control - Governed metric definition for AI Anomaly Acceptance.

P17-KPI-045 - AI Optimization Acceptance - Site Models & Job Files - Governed metric definition for AI Optimization Acceptance.

P17-KPI-046 - AI Evidence Coverage - Field Data Sync - Governed metric definition for AI Evidence Coverage.

P17-KPI-047 - Machine Control Adoption - Machine Telemetry Integration - Governed metric definition for Machine Control Adoption.

P17-KPI-048 - Machine Control Productivity Gain - Operator Interface - Governed metric definition for Machine Control Productivity Gain.

P17-KPI-049 - Machine Control Rework Reduction - Safety & Exclusion Controls - Governed metric definition for Machine Control Rework Reduction.

P17-KPI-050 - Machine Control Fuel/Productivity Impact - Design Change Management - Governed metric definition for Machine Control Fuel/Productivity Impact.

P17-KPI-051 - Machine Control Digital Twin Freshness - As-Built & Production Capture - Governed metric definition for Machine Control Digital Twin Freshness.

Engine Registry

P17-ENG-001 - Phase 16 Handoff Validation Engine - Deterministic - Provides phase 16 handoff validation engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-002 - Model Package Engine - Deterministic - Provides model package engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-003 - Model Revision Engine - Deterministic - Provides model revision engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-004 - Localization Validation Engine - Deterministic - Provides localization validation engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-005 - Localization Residual Engine - Deterministic - Provides localization residual engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-006 - CRS/Datum Compatibility Engine - Deterministic - Provides crs/datum compatibility engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-007 - Device Compatibility Engine - Deterministic - Provides device compatibility engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-008 - Attachment Compatibility Engine - Deterministic - Provides attachment compatibility engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-009 - Firmware Compatibility Engine - Deterministic - Provides firmware compatibility engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-010 - Job File Conversion Engine - Deterministic - Provides job file conversion engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-011 - Job File Packaging Engine - Deterministic - Provides job file packaging engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-012 - Checksum/Integrity Engine - Deterministic - Provides checksum/integrity engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-013 - Machine Assignment Engine - Deterministic - Provides machine assignment engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-014 - Operator Authorization Engine - Deterministic - Provides operator authorization engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-015 - Position Quality Engine - Deterministic - Provides position quality engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-016 - GNSS Health Engine - Deterministic - Provides gnss health engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-017 - Correction Service Health Engine - Deterministic - Provides correction service health engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-018 - 2D Guidance Engine - Deterministic - Provides 2d guidance engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-019 - 3D Guidance Engine - Deterministic - Provides 3d guidance engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-020 - Cut/Fill Engine - Deterministic - Provides cut/fill engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-021 - Grade/Slope Engine - Deterministic - Provides grade/slope engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-022 - Excavator Geometry Engine - Deterministic - Provides excavator geometry engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-023 - Dozer Blade Engine - Deterministic - Provides dozer blade engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-024 - Grader Cross-Slope Engine - Deterministic - Provides grader cross-slope engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-025 - Compactor Coverage Engine - Deterministic - Provides compactor coverage engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-026 - Compaction Pass Engine - Deterministic - Provides compaction pass engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-027 - Paving Guidance Engine - Deterministic - Provides paving guidance engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-028 - Milling Depth Engine - Deterministic - Provides milling depth engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-029 - Trench/Invert Engine - Deterministic - Provides trench/invert engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-030 - Station/Offset Engine - Deterministic - Provides station/offset engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-031 - Geofence Engine - Deterministic - Provides geofence engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-032 - No-Go Zone Engine - Deterministic - Provides no-go zone engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-033 - Safety Alert Engine - Deterministic - Provides safety alert engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-034 - Tolerance Engine - Deterministic - Provides tolerance engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-035 - Conformance Engine - Deterministic - Provides conformance engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-036 - As-Built Engine - Deterministic - Provides as-built engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-037 - Production Capture Engine - Deterministic - Provides production capture engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-038 - Material Movement Engine - Deterministic - Provides material movement engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-039 - Sync Engine - Deterministic - Provides sync engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-040 - Offline Conflict Engine - Deterministic - Provides offline conflict engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-041 - Transfer Verification Engine - Deterministic - Provides transfer verification engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-042 - Design Change Impact Engine - Deterministic - Provides design change impact engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-043 - Withdrawal/Reissue Engine - Deterministic - Provides withdrawal/reissue engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-044 - Telemetry Correlation Engine - Deterministic - Provides telemetry correlation engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-045 - Remote Diagnostics Engine - Deterministic - Provides remote diagnostics engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-046 - Device Health Engine - Deterministic - Provides device health engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-047 - Field Automation Rules Engine - Deterministic - Provides field automation rules engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-048 - Task Sequencing Engine - Deterministic - Provides task sequencing engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-049 - Model QA Engine - AI-assisted - Provides model qa engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-050 - Anomaly Detection Engine - AI-assisted - Provides anomaly detection engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-051 - Optimization Recommendation Engine - AI-assisted - Provides optimization recommendation engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-052 - Evidence Retrieval Engine - AI-assisted - Provides evidence retrieval engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-053 - AI Confidence Engine - AI-assisted - Provides ai confidence engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-054 - Machine Control KPI Engine - AI-assisted - Provides machine control kpi engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

P17-ENG-055 - Machine Control Digital Twin Engine - AI-assisted - Provides machine control digital twin engine using approved models/localization, machine/device context, telemetry, safety controls and lineage.

Workflow Registry

P17-WF-001 - Phase 16 Handoff Intake - Machine Control Core - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-002 - Machine-Control Project Setup - GPS/GNSS Positioning - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-003 - Localization Import & Review - Localization & Calibration - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-004 - Localization Field Verification - 3D Models & Surfaces - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-005 - Design Model Import - 2D Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-006 - Surface Validation - 3D Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-007 - Alignment/Corridor Validation - Automatic Grade Control - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-008 - Utility Model Validation - Excavator Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-009 - Job Package Creation - Dozer Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-010 - Job Package Approval - Grader Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-011 - Machine Assignment - Compactor Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-012 - Operator Authorization - Loader & Material Handling - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-013 - Device Compatibility Check - Paver/Milling Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-014 - Machine Job Load - Trenching & Utilities - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-015 - Operator Revision Acknowledgement - Road & Highway Machine Control - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-016 - 2D Guidance Session - Site Models & Job Files - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-017 - 3D Guidance Session - Field Data Sync - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-018 - Automatic Grade Enablement - Machine Telemetry Integration - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-019 - Manual Override - Operator Interface - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-020 - Excavator Guidance - Safety & Exclusion Controls - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-021 - Dozer Guidance - Design Change Management - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-022 - Grader Guidance - As-Built & Production Capture - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-023 - Compactor Guidance - Quality Assurance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-024 - Loader Guidance - Field Automation - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-025 - Paver Guidance - Remote Support & Diagnostics - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-026 - Milling Guidance - Device & Firmware Management - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-027 - Trench/Utility Guidance - Vendor/Platform Interoperability - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-028 - Road/Highway Guidance - AI Machine Intelligence - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-029 - Geofence/No-Go Publication - Machine Control Digital Twin - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-030 - Safety Alert & Escalation - Machine Control Core - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-031 - Position Quality Exception - GPS/GNSS Positioning - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-032 - Calibration Exception - Localization & Calibration - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-033 - Device Health Exception - 3D Models & Surfaces - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-034 - Field Issue Capture - 2D Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-035 - Design Change Impact Review - 3D Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-036 - Job Withdrawal - Automatic Grade Control - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-037 - Job Reissue - Excavator Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-038 - Machine-to-Cloud Sync - Dozer Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-039 - Cloud-to-Machine Sync - Grader Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-040 - Offline Conflict Resolution - Compactor Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-041 - As-Built Capture - Loader & Material Handling - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-042 - Production Capture - Paver/Milling Guidance - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-043 - Tolerance/Conformance Review - Trenching & Utilities - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-044 - Compaction QA Review - Road & Highway Machine Control - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-045 - Remote Support Session - Site Models & Job Files - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-046 - Firmware/Software Update Review - Field Data Sync - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-047 - Vendor File Conversion - Machine Telemetry Integration - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-048 - Machine Control QA Audit - Operator Interface - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-049 - AI Model/Anomaly Review - Safety & Exclusion Controls - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-050 - Machine Control Digital Twin Refresh - Design Change Management - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

P17-WF-051 - Closeout & Archive - As-Built & Production Capture - Initiate > Validate > Approve/Authorize > Deploy/Execute > Monitor > Verify > Acknowledge/Close

Machine-Control Governance

Approved models/localization, device compatibility, operator authority, safety boundaries and revision acknowledgement are mandatory controls. AI cannot autonomously enable machine control or override safety-critical limits.